



Spiking Neural Network Architecture & Utilization on FPGA

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1 Critical Path Calculation

1.1 Hardware Delay Metrics & Assumptions

To maximize throughput and minimize routing congestion, the architecture leverages the native Multiply-Accumulate (MAC) and 3-input adder capabilities of the UltraScale+ DSP48E2 slices.

- **DSP Cascade:** Operations are consolidated into cascaded DSP chains where 1 DSP handles both multiplication and accumulation (1 MAC = 1 Delay equivalent).
- **Conv25 & Conv9 Units:** Spatial convolutions are handled by dedicated cascaded DSP chains. The conv25 unit (Stage 1) has a critical path depth of **10 DSPs**, while the conv9 unit (Stages 2 & 3) has a depth of **9 DSPs**.

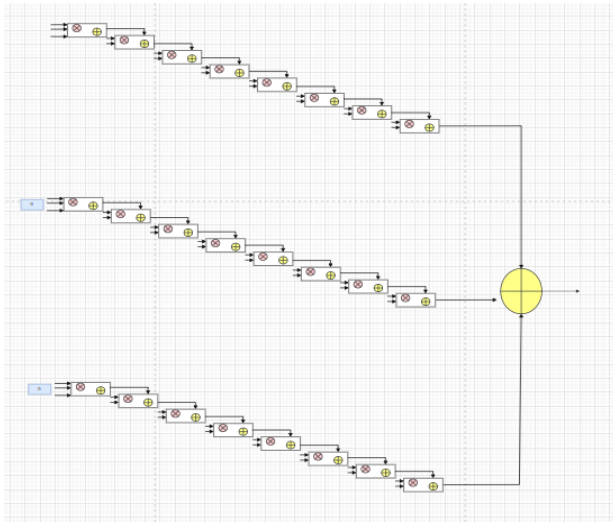


Figure 1: Conv25

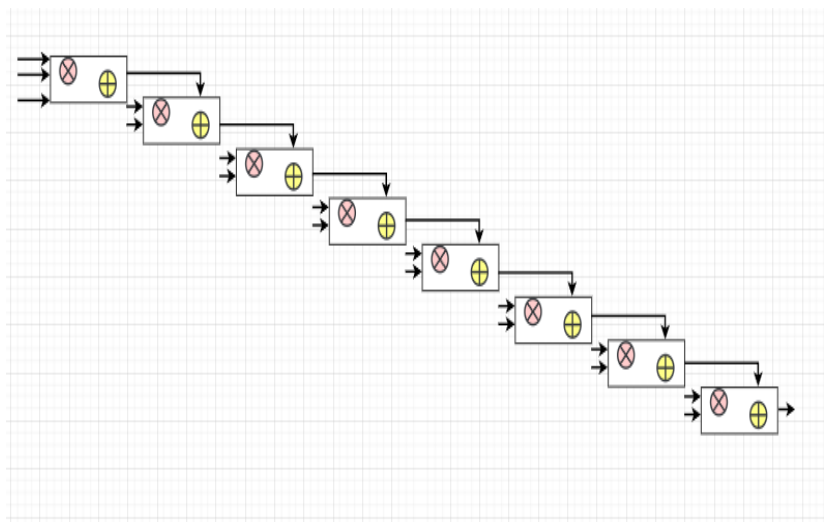


Figure 2: Conv9

- **Optimized Adder Trees:** Channel additions are optimized using 3-input adders, significantly reducing tree depth. Stage 2 requires a depth of **4 DSPs**, and Stage 3 requires a depth of **5 DSPs**.
- **Shaaban Unit:** The backend processing (Bias, BN, Pooling, and SNN/LIF logic) is unified into a single hardware block with an optimized critical path depth of **6 DSPs**.

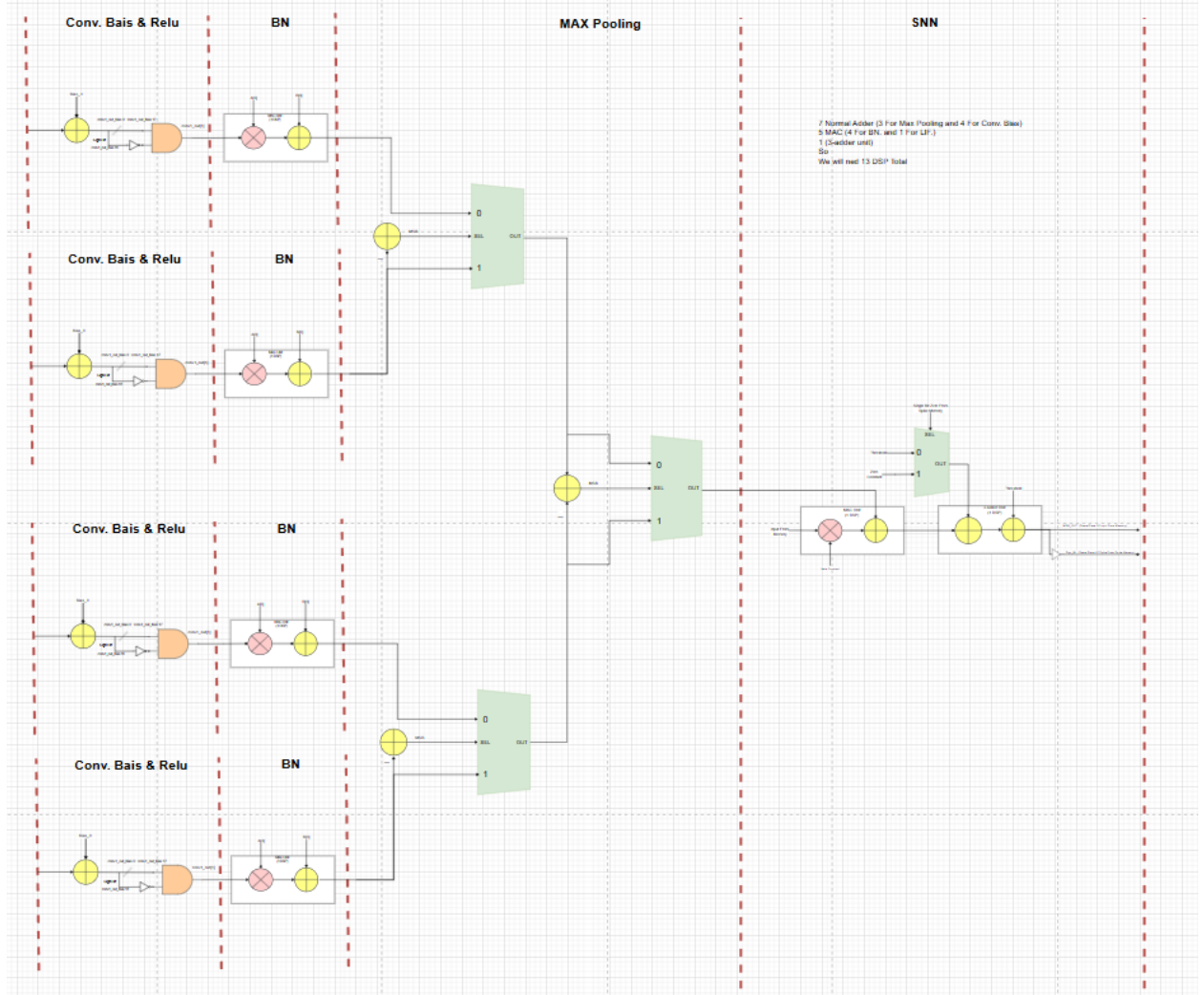


Figure 3: Shaaban unit

1.2 Pipeline Stage Critical Paths

Stage 1 Critical Path

- Spatial Conv (conv25 Unit): 10 delays
- Backend Processing (Shaaban Unit): 6 delays
- Total Stage 1 Critical Path: $10 + 6 = 16$ delays

Stage 2 Critical Path

- Spatial Conv (conv9 Unit): 9 delays
- 3-Input Adder Tree (Stage 2): 4 delays
- Backend Processing (Shaaban Unit): 6 delays
- Total Stage 2 Critical Path: $9 + 4 + 6 = 19$ delays



Stage 3 Critical Path

- Spatial Conv (conv9 Unit): 9 delays
- 3-Input Adder Tree (Stage 3): 5 delays
- Backend Processing (Shaaban Unit): 6 delays
- Total Stage 3 Critical Path: $9 + 5 + 6 = 20$ delays

Summary Note: The maximum operating frequency of the entire network is bounded by Stage 3 at a critical path of **20 delays**

2 Hardware Resource Allocation & Cycle Documentation

2.1 Global Hardware Constraints

The total available computational footprint on the FPGA is capped at 4,638 DSP Blocks.

A critical optimization in this architecture is the physical instantiation and sharing of the backend Shaaban Units. Exactly 32 Shaaban units (costing 13 DSPs each, or 416 total DSPs) are physically instantiated on the FPGA:

- Stage 1: Actively utilizes all 32 units (29 dedicated + 2 shared with Stage 2 + 1 shared across all stages).
- Stage 2: Actively utilizes 3 units (2 shared + 1 shared). The remaining 29 units sit idle but are accounted for as static hardware overhead.
- Stage 3: Actively utilizes 1 unit (1 shared). The remaining 31 units sit idle but are accounted for as static hardware overhead.

2.2 SNN Stages Resource and Cycle Summary

The following table summarizes the component-level DSP consumption and the resulting spatial throughput (pixels per cycle) for each pipeline stage.

Architectural Component	Stage 1	Stage 2	Stage 3
Output Throughput	32 pixels / cycle	3 pixels / cycle	1 pixel / cycle
Convolution Units	128 conv25 units	384 conv9 units	256 conv9 units
Conv DSP Count	$128 \times 27 = 3,456$ DSPs	$384 \times 9 = 3,456$ DSPs	$256 \times 9 = 2,304$ DSPs
Adder Tree DSP Count	$10 \times 4 \times 3 = 120$ DSPs	$16 \times 4 \times 3 = 192$ DSPs	$33 \times 4 = 132$ DSPs
Extra ADD (conv25)	8 DSPs	N/A	N/A
Shaaban Unit Overhead	416 DSPs (32 units)	39 DSPs (3 Only used)	13 DSPs (1 Only used)
Total DSPs Used (Utilization)	4,000 DSPs (98.1%)	3,687 DSPs (90.45%)	2,449 DSPs (60.08%)

So total number of DSPs used in whole circuit = $4000 + (4+1+1) \times 12 + 4 = 4076$ DSPs



3 Total Processing Time (Latency) Calculation

To determine the total real-world latency required to process a full batch of 16 image frames, we calculate the total pipeline cycles across the defined spatial resolution and divide it by the hardware's operating frequency.

System Parameters:

- Cycles per pixel block: 100 (Stage 1) + 352 (Stage 2) + 128 (Stage 3) = 580 cycles
- Spatial Resolution: 32x32 pixels (1,024 pixels per frame)
- Number of Frames: 16 frames
- Hardware Operating Frequency: 40 MHz (40,000,000 cycles per second)

Time Calculation:

- Total workload cycles: $580 \text{ cycles/pixel} \times 1,024 \text{ pixels/frame} \times 16 \text{ frames} = 9,502,720 \text{ cycles}$
- Total Processing Time: $9,502,720 \text{ cycles} / 40,000,000 \text{ Hz} = 0.237568 \text{ seconds}$